Routing Drift in Neural-Collapse Continual Learning: Supplementary Material

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A Supplement Overview

This supplement is organized as a companion to the main paper. We first record the implementation details needed to reproduce the trained artifacts, including the reference ProNC objective and hyperparameters. We then collect notation and neural-collapse background, and define the classifier variants evaluated from the same frozen features and saved task artifacts. The remaining sections separate three questions: which variant is strongest, which components matter, and whether the observed gains match the routing-drift mechanism. Throughout, RCR denotes the deployed Reliability-Calibrated Routing score; other heads are controlled variants or ablations.

B Implementation Details

This section records the training and evaluation details behind the frozen artifacts used in the main paper. Our contribution is evaluated as a classifier correction on top of a trained progressive ETF model; the backbone checkpoints are produced with the reference ProNC objective [2], while RCR is applied after training from saved task artifacts.

Reference training objective. The first task is trained with ordinary supervised cross-entropy. At later tasks, the ETF frame is expanded before learning the new task, and each update uses a minibatch containing current-task samples and replay samples. Let $z_t(x) = f_{\theta_t}(x)/\|f_{\theta_t}(x)\|$ be the normalized current feature, $z_{t-1}(x)$ the normalized feature from the frozen previous model, and $u_{y,t}$ the progressive ETF vertex assigned to label y at stage t. The training loss is

$$\mathcal{L}_{\text{ProNC}} = \mathcal{L}_{\text{ce}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{distill}}. \tag{1}$$

The alignment term is the squared cosine distance-to-one objective

$$\mathcal{L}_{\text{align}}(x,y) = \frac{1}{2}\left(\langle z_t(x), u_{y,t}\rangle - 1\right)^2, \tag{2}$$

which pulls the current normalized feature toward the ETF vertex of its class. The feature-distillation term is

$$\mathcal{L}_{\text{distill}}(x) = \frac{1}{2}\left(\langle z_t(x), z_{t-1}(x)\rangle - 1\right)^2, \tag{3}$$

Table 1: **Training hyperparameters for reproduced artifacts.** The same checkpoints are used to evaluate both the reference ETF head and RCR in paired comparisons.

Dataset	Buffer	Epochs	LR	Milestones	λ_1	λ_2
Seq-CIFAR-10	200	50	0.01	35, 45	13	90
Seq-CIFAR-10	500	50	0.01	35, 45	12	80
Seq-CIFAR-100	200	50	0.03	35, 45	18	170
Seq-CIFAR-100	500	50	0.03	35, 45	18	170
Seq-TinyImageNet	200	100	0.03	45, 75, 95	20	165
Seq-TinyImageNet	500	100	0.03	45, 75, 95	20	165

evaluated on the same mixed minibatch. Thus, λ_1 controls the strength of ETF alignment and λ_2 controls the strength of feature preservation from the previous model. The cross-entropy term keeps its unit coefficient in all reproduced runs.

Training hyperparameters. All reproduced artifacts use a ResNet-18 backbone, batch size 32, SGD with zero momentum and zero weight decay, and a multistep learning-rate schedule with decay factor 0.1. The class order is fixed by the benchmark split, and all reported paired results average seeds $0,1,2$. The dataset-specific values used to produce the saved checkpoints are listed in Table 1.

Saved artifacts and post-hoc evaluation. At each task boundary we save the checkpoint, the expanded ETF frame, replay features needed for class representatives, and the previous-to-current feature transport fit. The reference ETF head predicts with

$$\hat{y}_{\mathrm{ETF}}(x) = \arg\max_{c \in \mathcal{C}_t} \langle z_t(x), u_{c,t} \rangle . \tag{4}$$

RCR uses the same frozen feature $z_t(x)$ and the same ETF frame. It adds only the transported class representative, competitive radius, NC retraction, and fixed ETF/radial composition defined in the main paper. No scalar is chosen per dataset or from a validation split: all datasets use $\alpha_\rho = 0.25$, $\gamma = 0.5$, shrink clipping to $[0, 0.95]$, radius clipping to $[10^{-4}, 10]$, and $\omega = q/(q+1)$.

Transport fitting. Class representatives are moved across feature frames by fitting a ridge linear map between previous-model and current-model normalized features from the mixed training stream at a task boundary. The reported artifacts use ridge coefficient 10^{-3}, pair normalization before fitting, and normalized features at evaluation. The transported representative p_c is therefore a class-level measure in the current feature frame, while u_c remains the global ETF vertex used by the neural-collapse classifier.

C Notation and Neural-Collapse Background

Table 2 summarizes the notation used throughout the paper and this supplement. We keep protocol symbols, routing diagnostics, and RCR components separate to avoid overloading task indices, class sets, and prototype quantities.

Table 2: **Primary notation.** The first column makes each object's type explicit. We use calligraphic letters for sets, lower-case symbols for feature/prototype vectors and scalars, upper-case Roman letters for matrices, and $\mathcal{L}_{\text{name}}$ for losses.

Type	Symbol	Description
Protocol and indexing		
Set	$\mathcal{T}=\{1,\dots,T\}$	Ordered continual-learning task sequence.
Index	t	Current training or evaluation stage.
Set	$\mathcal{D}_t$	Training data introduced by task t.
Set	$\mathcal{C}_t$	Classes observed after task t.
Scalar	q	Number of classes per task block.
Function	$\tau(c)$	Task block containing class c.
Set	$\mathcal{B}_c$	Replay samples stored for class c.
Metric	FAA, FF	Final average accuracy and final forgetting.
Features, vectors, and matrices		
Function	f_θ	Current feature extractor.
Vector	$z(x)$	Normalized current feature, $f_\theta(x)/\|f_\theta(x)\|$.
Vector	$z^-(x)$	Normalized feature from the previous model.
Vector	u_c	Progressive ETF vertex for class c.
Matrix	A_t	Ridge previous-to-current feature transport at stage t.
Vector	p_c	Transported class representative in the current feature frame.
Vector	$\tilde{p}_c$	NC-shrunk class representative used by the radial score.
Operator	$d_{\cos}(a,b)$	Cosine-distance convention $2-2\langle a,b\rangle$ for normalized vectors.
Operator	$\mathcal{N}[\cdot]$	Per-sample standardization over currently seen classes.
Scores, radii, and scalars		
Score	$s_{\text{ETF}}(x,c)$	ETF cosine score for feature $z(x)$ and vertex u_c.
Score	$s_{\text{rad}}(x,c)$	Radius-normalized radial score around the NC-shrunk class representative.
Score	$s_{\text{RCR}}(x,c)$	Final RCR score combining standardized ETF and radial evidence.
Scalar	r_c	Raw class radius estimated from replay features.
Scalar	$\bar{r}$	Mean raw radius over currently seen classes.
Scalar	ρ_c	Competitive class-dependent reliability radius.
Scalar	$\bar{\rho}$	Mean competitive radius.
Scalar	α_ρ	Radius-compression exponent used to form competitive radii.
Scalar	α_c	NC-shrink coefficient used to retract p_c toward u_c.
Scalar	γ	Exponent controlling the shrink coefficient.
Scalar	ω	Fixed ETF/radial mixture weight, $\omega=q/(q+1)$.
Scalar	ε	Numerical stability constant.
Losses and routing diagnostics		
Loss	$\mathcal{L}_{\text{ProNC}}$	Reference ProNC training objective.
Loss	$\mathcal{L}_{\text{ce}}$	Supervised cross-entropy term.
Loss	$\mathcal{L}_{\text{align}}$	ETF-alignment term in the reference objective.
Loss	$\mathcal{L}_{\text{distill}}$	Feature-distillation term from the previous model.
Score	$m_b(x)$	Maximum class score inside task block b.
Scalar	$\Delta_{\text{route}}(x)$	True-block score margin, $m_{\tau(y)}(x)-\max_{b\neq\tau(y)} m_b(x)$.
Event	$R(x)$	Indicator that the global argmax lies in the correct task block.
Event	$Q(x)$	Indicator that the correct class wins inside the true block.
Metric	Route accuracy	Test-set expectation of $R(x)$.

We recall the standard neural-collapse terminology used in the main paper. Consider a balanced K-class classification problem. Let $h_{k,i}\in\mathbb{R}^d$ be the last-layer feature of sample i from class k, μ_k the class mean, $\mu_G=\frac{1}{K}\sum_{k=1}^{K}\mu_k$ the global mean, and w_k the classifier weight for class k. Neural collapse describes the terminal training regime in which the following four properties emerge [1].

NC1: within-class variability collapse. Features from the same class concentrate around their class mean:

$$\Sigma_W=\frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_i(h_{k,i}-\mu_k)(h_{k,i}-\mu_k)^\top\to 0. \tag{5}$$

NC2: simplex ETF structure of class means. The centered class means

$$\tilde{\mu}_k = \frac{\mu_k - \mu_G}{\|\mu_k - \mu_G\|} \tag{6}$$

approach a simplex equiangular tight frame, so all classes have equal norm and equal pairwise angle:

$$\langle \tilde{\mu}_i, \tilde{\mu}_j \rangle = -\frac{1}{K-1} \qquad i \neq j. \tag{7}$$

NC3: self-duality. Classifier weights align with centered class means:

$$\frac{w_k}{\|w_k\|} \approx \tilde{\mu}_k. \tag{8}$$

NC4: nearest-class-center inference. Under NC1–NC3, the classifier decision is equivalent to nearest-class-center or nearest-prototype inference in the collapsed feature space. This is the connection exploited by ETF classifiers: once the class directions form a simplex ETF, prediction can be performed by comparing a normalized feature to class vertices through cosine similarity.

D Offline Classifier Variants

All variants below are evaluated on the same frozen feature tensor for a given run. We use z for the normalized feature, u_c for the ProNC ETF vertex of class c, and p_c for the transported class prototype in the current feature frame:

$$z = \frac{f_\theta(x)}{\|f_\theta(x)\|}. \tag{9}$$

We write $\mathcal{N}[\cdot]$ for per-sample standardization across the currently seen classes. Unless stated otherwise, no variant uses test-time task identity or a learned task oracle.

Evaluation artifacts. All variants are evaluated from saved task and feature artifacts. The same saved features are used for ProNC ETF, LDC-NCM, radial scoring, and reliability-calibrated scoring. This isolates the classifier effect from training variance.

ETF reference. This is the nearest-ETF score used by the base progressive neural-collapse classifier:

$$s_{\mathrm{ETF}}(x,c) = \langle z, u_c \rangle. \tag{10}$$

This is the baseline head in the main tables.

CE head (audit). The trained linear classifier head uses its saved weights and bias. We report it only as an audit row, because the controlled setting evaluates neural-collapse heads with ETF inference rather than this linear classifier.

Transported NCM. This variant uses transported nearest-centroid inference:

$$s_{\mathrm{LDC}}(x,c) = \langle z, p_c \rangle\,. \tag{11}$$

It tests whether moving class representatives into the current feature frame is useful without adding radial reliability.

Radial-only. This variant scores local plausibility by normalizing the cosine distance to p_c with a class radius:

$$s_{\mathrm{rad}}(x,c) = -\frac{2-2\langle z, p_c \rangle}{\rho_c + \varepsilon}. \tag{12}$$

It tests whether local class reliability alone can repair cross-task routing.

CompMix. CompMix composes the ETF score with the competitive-radius radial score around the transported prototype:

$$s_{\mathrm{CompMix}}(x,c) = (1-\lambda_t)\mathcal{N}[s_{\mathrm{ETF}}](x,c) + \lambda_t \mathcal{N}[s_{\mathrm{rad}}](x,c), \qquad \lambda_t = \min\left(\frac{q}{q+1}, \frac{t-1}{t}\right). \tag{13}$$

The deterministic stage schedule keeps the radial term weak at early tasks and approaches the fixed cap as more task blocks are observed.

CompMix-Fixed. CompMix-Fixed uses the same competitive-radius radial score around p_c, but removes the stage schedule and uses the fixed cap $\omega = q/(q+1)$ at every task:

$$s_{\mathrm{CompMixFix}}(x,c) = (1-\omega)\mathcal{N}[s_{\mathrm{ETF}}](x,c) + \omega \mathcal{N}[s_{\mathrm{rad}}](x,c). \tag{14}$$

It isolates whether the gain comes from the evidence composition itself or from a deterministic task-stage schedule.

InvGradMix (directional sensitivity). InvGradMix replaces the Euclidean radial distance with a global diagonal metric $M_t = \mathrm{diag}(m)$ estimated from replay samples using inverse empirical CE-gradient magnitude:

$$s_{\mathrm{invgrad}}(x,c) = -\frac{(z-p_c)^\top M_t (z-p_c)}{\rho_c + \varepsilon}. \tag{15}$$

It then uses the same stage schedule λ_t as CompMix. This variant tests whether different feature directions should have different reliability.

InvGradMix-Fixed (directional sensitivity). InvGradMix-Fixed uses the same direction-aware radial score as InvGradMix, but uses the fixed cap $\omega = q/(q+1)$ instead of the stage schedule. It is a stronger direction-aware variant, but it introduces metric-estimation choices that are not needed by RCR.

NC-ShrinkMix. NC-ShrinkMix adds the neural-collapse retraction to CompMix. It first retracts the transported prototype toward the ETF vertex,

$$\tilde{p}_c = \mathrm{norm}\left(\alpha_c p_c + (1-\alpha_c)u_c\right), \qquad \alpha_c = \mathrm{clip}\left[\left(\frac{\bar{\rho}}{\rho_c+\varepsilon}\right)^{\gamma}, 0, 0.95\right], \tag{16}$$

with $\gamma = 0.5$. It then recomputes the competitive class radius ρ_c and combines ETF evidence with radial evidence using the scheduled mixture λ_t.

RCR. RCR is the fixed-cap version of NC-ShrinkMix. It uses the retracted prototype $\tilde{p}_c$, recomputes the competitive radius around that prototype, and combines ETF evidence with radial evidence using $\omega = q/(q+1)$:

$$s_{\mathrm{RCR}}(x,c) = (1-\omega)\mathcal{N}[s_{\mathrm{ETF}}](x,c) + \omega\mathcal{N}[s_{\mathrm{rad}}](x,c). \tag{17}$$

This is the method used in all main-paper results.

Saved mix (audit). Some artifacts contain the mixture that was active when the run was produced. Because this score can depend on historical choices of auxiliary head, radius mode, and mixture weight, we report it only as an audit row. It is not a method candidate and is included only to make older saved artifacts auditable.

No class-index prior. The proposed score does not use the numerical class index, task age, or a hand-coded penalty such as $a_{\tau(c)}$ or bc in the logits. The only mixture quantity in RCR is the global scalar $\omega = q/(q+1)$, shared by every class and sample. Class-specific terms are data-derived representatives and uncertainties, (u_c, p_c, ρ_c), exactly as in prototype-based classifiers. The task index is used for evaluation metrics such as Task-IL and route accuracy, but not as a test-time oracle or as a per-class routing bias.

E Classifier Variant Benchmark

The main paper reports the paired ETF-versus-RCR comparison. Here we show the broader classifier family on the same frozen artifacts. The purpose is not to introduce multiple methods, but to verify which ingredient explains the gains. RCR is the deployed variant because it combines ETF evidence, transported class reliability, and NC retraction without the additional metric-estimation choices used by the direction-aware variants. Rows explicitly marked as audits are diagnostic controls rather than candidate methods; they are included to show which apparent shortcuts fail under the same frozen-feature evaluation.

Analysis. Table 3 shows three consistent patterns. First, the audit CE head is not a competitive replacement for ETF inference: it often preserves Task-IL but collapses Class-IL, confirming that the comparison should be made between neural-collapse heads on the same feature artifacts. Second, transported NCM and radial-only scoring are useful but incomplete: they improve hard routing regimes such as Seq-CIFAR-100 and Seq-TinyImageNet, yet either underperform on benign Seq-CIFAR-10 or lose the global ETF ordering. Third, the large gains appear when the ETF score and local radial reliability are composed. CompMix and NC-ShrinkMix therefore separate the two useful signals: global ETF comparability and local class plausibility. The fixed variants remove the deterministic stage schedule; RCR is selected because it gives the most stable forgetting reduction while keeping the isotropic NC story intact. The direction-aware InvGradMix variants are deliberately left as ablations: they show that additional feature-direction reliability can sometimes improve FAA or Route, but at the cost of an extra learned metric and a less parsimonious method.

Table 3: **Classifier variants evaluated on frozen artifacts.** All variants use the same feature artifacts and test split. Route is final route accuracy.

Dataset	Buffer	Head	n	Class FAA ↑	Class FF ↓	Task FAA ↑	Task FF ↓	Route ↑
		ETF reference	3	66.14 ± 1.69	31.47 ± 2.46	**96.74 ± 0.12**	**0.62 ± 0.20**	66.58 ± 1.64
		CE head (audit)	3	54.63 ± 0.15	52.99 ± 0.03	96.62 ± 0.24	0.75 ± 0.24	55.04 ± 0.09
		Transported NCM	3	63.46 ± 0.75	20.75 ± 1.21	96.11 ± 0.53	1.36 ± 0.66	63.98 ± 0.66
		Radial-only	3	64.38 ± 1.23	**15.80 ± 1.84**	96.11 ± 0.53	1.36 ± 0.66	65.06 ± 0.98
		CompMix	3	69.84 ± 0.76	24.39 ± 1.40	96.63 ± 0.10	0.75 ± 0.22	70.30 ± 0.80
CIFAR-10	200	CompMix-Fixed	3	69.84 ± 0.76	24.15 ± 1.46	96.63 ± 0.10	0.77 ± 0.22	70.30 ± 0.80
		InvGradMix	3	**70.65 ± 0.71**	22.59 ± 1.17	96.72 ± 0.09	0.67 ± 0.14	**71.11 ± 0.73**
		InvGradMix-Fixed	3	70.65 ± 0.71	22.16 ± 1.18	96.72 ± 0.09	0.69 ± 0.13	71.11 ± 0.73
		NC-ShrinkMix	3	70.36 ± 0.62	23.03 ± 1.40	96.67 ± 0.09	0.72 ± 0.19	70.82 ± 0.56
		RCR	3	70.36 ± 0.62	22.85 ± 1.45	96.67 ± 0.09	0.73 ± 0.19	70.82 ± 0.56
		Saved mix (audit)	3	68.41 ± 0.58	26.72 ± 0.93	96.59 ± 0.30	0.80 ± 0.38	68.93 ± 0.51
		ETF reference	3	72.16 ± 1.59	24.18 ± 2.14	96.82 ± 0.14	**0.47 ± 0.04**	72.71 ± 1.65
		CE head (audit)	3	65.60 ± 0.48	38.90 ± 0.48	**96.86 ± 0.15**	0.51 ± 0.10	66.03 ± 0.48
		Transported NCM	3	66.28 ± 0.34	18.60 ± 0.53	96.34 ± 0.04	0.97 ± 0.13	66.95 ± 0.36
		Radial-only	3	67.81 ± 1.05	**12.52 ± 1.58**	96.34 ± 0.04	0.97 ± 0.13	68.51 ± 0.96
		CompMix	3	73.56 ± 0.04	18.42 ± 0.85	96.76 ± 0.01	0.59 ± 0.12	74.22 ± 0.19
CIFAR-10	500	CompMix-Fixed	3	73.56 ± 0.04	18.35 ± 0.82	96.76 ± 0.01	0.59 ± 0.13	74.22 ± 0.19
		InvGradMix	3	**74.00 ± 0.19**	17.25 ± 0.94	96.86 ± 0.09	0.48 ± 0.01	**74.65 ± 0.32**
		InvGradMix-Fixed	3	74.00 ± 0.19	17.06 ± 1.01	96.86 ± 0.09	0.49 ± 0.02	74.65 ± 0.32
		NC-ShrinkMix	3	73.68 ± 0.88	17.78 ± 0.90	96.81 ± 0.04	0.54 ± 0.07	74.33 ± 1.02
		RCR	3	73.68 ± 0.88	17.73 ± 0.88	96.81 ± 0.04	0.54 ± 0.08	74.33 ± 1.02
		Saved mix (audit)	3	72.90 ± 0.56	19.97 ± 0.76	96.79 ± 0.01	0.52 ± 0.15	73.53 ± 0.49
		ETF reference	3	41.71 ± 0.75	38.55 ± 1.07	85.53 ± 0.20	4.51 ± 0.40	42.81 ± 0.70
		CE head (audit)	3	19.77 ± 0.51	77.33 ± 0.88	86.32 ± 0.10	3.76 ± 0.20	20.48 ± 0.51
		Transported NCM	3	43.59 ± 0.79	31.24 ± 1.00	85.64 ± 0.09	4.03 ± 0.17	45.41 ± 0.78
		Radial-only	3	46.78 ± 0.79	21.47 ± 0.80	85.64 ± 0.09	4.03 ± 0.17	48.92 ± 0.65
		CompMix	3	49.28 ± 0.50	21.13 ± 0.46	87.04 ± 0.21	2.76 ± 0.29	50.99 ± 0.38
CIFAR-100	200	CompMix-Fixed	3	49.20 ± 0.48	18.24 ± 0.47	87.00 ± 0.18	2.62 ± 0.18	50.93 ± 0.37
		InvGradMix	3	49.55 ± 0.60	20.76 ± 0.68	87.17 ± 0.22	2.57 ± 0.17	51.26 ± 0.50
		InvGradMix-Fixed	3	49.46 ± 0.51	18.00 ± 0.50	87.13 ± 0.19	2.47 ± 0.17	51.21 ± 0.42
		NC-ShrinkMix	3	49.72 ± 0.66	19.31 ± 0.62	**87.37 ± 0.28**	2.45 ± 0.22	51.38 ± 0.50
		RCR	3	**49.74 ± 0.71**	**16.27 ± 0.45**	87.28 ± 0.25	**2.36 ± 0.14**	**51.43 ± 0.57**
		Saved mix (audit)	3	44.53 ± 5.92	33.58 ± 16.01	87.15 ± 0.36	2.57 ± 0.12	46.01 ± 6.23
		ETF reference	3	48.65 ± 0.68	23.16 ± 1.51	85.54 ± 0.23	4.29 ± 0.31	50.14 ± 0.51
		CE head (audit)	3	28.56 ± 0.42	66.87 ± 0.55	86.27 ± 0.25	3.85 ± 0.11	29.39 ± 0.46
		Transported NCM	3	50.95 ± 0.31	21.25 ± 0.48	86.94 ± 0.19	2.63 ± 0.19	52.83 ± 0.22
		Radial-only	3	51.96 ± 0.06	13.84 ± 0.18	86.94 ± 0.19	2.63 ± 0.19	53.94 ± 0.07
		CompMix	3	53.15 ± 0.49	12.71 ± 0.26	87.55 ± 0.15	2.30 ± 0.05	54.95 ± 0.34
CIFAR-100	500	CompMix-Fixed	3	53.07 ± 0.41	11.44 ± 0.26	87.51 ± 0.15	2.19 ± 0.11	54.86 ± 0.24
		InvGradMix	3	**53.29 ± 0.40**	12.76 ± 0.30	87.58 ± 0.07	2.30 ± 0.09	**55.11 ± 0.25**
		InvGradMix-Fixed	3	53.24 ± 0.38	11.42 ± 0.24	87.59 ± 0.07	2.17 ± 0.13	55.07 ± 0.20
		NC-ShrinkMix	3	53.29 ± 0.49	11.32 ± 0.26	**87.76 ± 0.07**	2.14 ± 0.12	55.04 ± 0.32
		RCR	3	53.27 ± 0.41	**10.18 ± 0.38**	87.74 ± 0.07	2.02 ± 0.16	55.04 ± 0.26
		Saved mix (audit)	3	50.23 ± 4.17	24.12 ± 14.59	87.71 ± 0.08	**2.00 ± 0.25**	51.88 ± 4.37
		ETF reference	3	25.77 ± 0.50	43.45 ± 0.13	68.30 ± 0.76	9.58 ± 0.42	28.58 ± 0.38
		CE head (audit)	3	9.91 ± 0.09	75.27 ± 0.33	**71.98 ± 0.35**	**6.52 ± 0.09**	12.20 ± 0.10
		Transported NCM	3	28.71 ± 0.40	33.37 ± 0.45	68.69 ± 0.38	9.65 ± 0.40	32.17 ± 0.45
		Radial-only	3	30.46 ± 0.20	26.66 ± 0.72	68.69 ± 0.38	9.65 ± 0.40	33.99 ± 0.16
		CompMix	3	30.68 ± 0.78	30.89 ± 0.88	69.79 ± 0.99	8.35 ± 1.02	33.96 ± 0.68
TinyImageNet	200	CompMix-Fixed	3	30.18 ± 0.92	26.84 ± 1.29	68.54 ± 0.98	9.62 ± 1.03	33.74 ± 0.90
		InvGradMix	3	30.36 ± 0.89	31.20 ± 0.95	69.75 ± 0.90	8.42 ± 0.93	33.59 ± 0.87
		InvGradMix-Fixed	3	29.92 ± 0.91	27.41 ± 1.12	68.53 ± 0.95	9.76 ± 1.04	33.38 ± 0.86
		NC-ShrinkMix	3	**31.55 ± 0.72**	28.41 ± 0.83	70.55 ± 0.94	7.57 ± 0.88	34.70 ± 0.55
		RCR	3	31.42 ± 0.80	**23.84 ± 1.15**	69.59 ± 0.99	8.61 ± 1.09	**34.76 ± 0.68**
		Saved mix (audit)	3	31.24 ± 0.44	27.95 ± 0.54	70.23 ± 0.51	8.07 ± 0.48	34.53 ± 0.44
		ETF reference	3	26.61 ± 0.39	41.16 ± 0.23	67.71 ± 0.47	9.78 ± 0.52	29.37 ± 0.52
		CE head (audit)	3	10.81 ± 0.33	73.80 ± 0.38	**71.59 ± 0.44**	**6.69 ± 0.60**	13.22 ± 0.28
		Transported NCM	3	29.59 ± 0.46	32.10 ± 0.40	69.37 ± 0.43	8.71 ± 0.40	32.92 ± 0.29
		Radial-only	3	31.23 ± 0.08	25.38 ± 0.24	69.37 ± 0.43	8.71 ± 0.40	34.67 ± 0.02
		CompMix	3	31.92 ± 0.01	28.72 ± 0.35	70.32 ± 0.32	7.71 ± 0.37	35.16 ± 0.15
TinyImageNet	500	CompMix-Fixed	3	31.34 ± 0.15	25.20 ± 0.25	69.21 ± 0.20	8.91 ± 0.20	34.71 ± 0.19
		InvGradMix	3	31.68 ± 0.12	29.10 ± 0.53	70.47 ± 0.41	7.68 ± 0.50	34.91 ± 0.05
		InvGradMix-Fixed	3	31.09 ± 0.21	25.78 ± 0.18	69.32 ± 0.18	8.87 ± 0.17	34.44 ± 0.11
		NC-ShrinkMix	3	**32.47 ± 0.17**	26.67 ± 0.23	70.89 ± 0.44	7.10 ± 0.56	**35.63 ± 0.27**
		RCR	3	32.24 ± 0.05	**22.47 ± 0.13**	70.16 ± 0.20	7.90 ± 0.16	35.46 ± 0.08
		Saved mix (audit)	3	31.93 ± 0.13	26.80 ± 0.36	70.53 ± 0.42	7.48 ± 0.40	35.29 ± 0.12

Note: Bold marks the best mean and underline the second best mean within each dataset/buffer block. Saved mix is included as an audit row because it depends on the score stored at artifact generation time.

Table 4: **Diagnostic check on a second progressive-ETF learner.** We reproduce an NCT-style progressive ETF run on Seq-CIFAR-100 with buffer 200 and evaluate an offline RCR-style ramp correction on the same saved features. The ramp keeps ETF evidence dominant at early tasks and increases the radial correction later. The purpose is diagnostic rather than a full re-benchmarking of NCT: it checks that the route-correction signal is not unique to the ProNC implementation.

Head	Class FAA ↑	Class FF ↓	Task FAA ↑	Task FF ↓	Route ↑
NCT-style ETF	31.62	27.05	79.93	4.99	33.58
NCT-style ETF + RCR-style ramp	34.02	25.84	81.33	3.31	35.89
Δ	+2.40	−1.21	+1.41	−1.68	+2.32

F Second Progressive-ETF Diagnostic

Reviewer feedback asked whether the route-correction mechanism is tied to one progressive-ETF implementation. To address this without overclaiming, we ran a small diagnostic on an NCT-style progressive ETF model and applied an offline RCR-style ramp correction to the same frozen features. This is not intended as a full NCT reproduction or a replacement for the controlled ProNC comparison in the main paper. It simply checks whether the same route-repair signal appears when the base learner is changed within the NC/ETF family.

G Component and Sensitivity Ablations

G.1 Score Components

Table 5: **Transport, radial reliability, and NC shrinkage are complementary.**

Dataset/Buffer	Head	Class-IL ↑	Task-IL ↑	Route ↑
Seq-CIFAR-10 / 200	ETF only	66.14 ± 1.38	96.74 ± 0.10	66.58 ± 1.34
	Transported NCM	63.46 ± 0.61	96.11 ± 0.43	63.98 ± 0.54
	Radial only	64.06 ± 0.42	96.27 ± 0.15	64.66 ± 0.32
	ETF+NCM	66.31 ± 0.82	96.55 ± 0.26	66.80 ± 0.70
	ETF+raw radial	68.50 ± 0.33	96.62 ± 0.08	69.05 ± 0.29
	ETF+equal radius	66.31 ± 0.82	96.55 ± 0.26	66.80 ± 0.70
	ETF+no transport	66.14 ± 1.38	96.76 ± 0.09	66.58 ± 1.34
	Ours (RCR)	70.36 ± 0.50	96.67 ± 0.08	70.82 ± 0.46
Seq-CIFAR-10 / 500	ETF only	72.16 ± 1.30	96.82 ± 0.12	72.71 ± 1.35
	Transported NCM	66.28 ± 0.27	96.34 ± 0.03	66.95 ± 0.29
	Radial only	68.90 ± 0.81	96.24 ± 0.02	69.65 ± 0.69
	ETF+NCM	71.73 ± 0.69	96.72 ± 0.04	72.32 ± 0.61
	ETF+raw radial	71.90 ± 1.64	96.73 ± 0.13	72.61 ± 1.65
	ETF+equal radius	71.73 ± 0.69	96.72 ± 0.04	72.32 ± 0.61
	ETF+no transport	72.16 ± 1.30	96.83 ± 0.11	72.71 ± 1.35
	Ours (RCR)	73.68 ± 0.72	96.81 ± 0.03	74.33 ± 0.83
Seq-CIFAR-100 / 200	ETF only	41.71 ± 0.61	85.53 ± 0.16	42.81 ± 0.57
	Transported NCM	43.59 ± 0.65	85.64 ± 0.07	45.41 ± 0.64
	Radial only	46.21 ± 0.44	85.46 ± 0.26	48.36 ± 0.49
	ETF+NCM	44.49 ± 0.29	86.64 ± 0.18	46.04 ± 0.28
	ETF+raw radial	43.10 ± 1.07	86.57 ± 0.42	44.87 ± 1.05
	ETF+equal radius	44.49 ± 0.29	86.64 ± 0.18	46.04 ± 0.28
	ETF+no transport	41.71 ± 0.61	85.56 ± 0.16	42.81 ± 0.57
	Ours (RCR)	49.74 ± 0.58	87.28 ± 0.20	51.43 ± 0.47
Seq-CIFAR-100 / 500	ETF only	48.65 ± 0.55	85.54 ± 0.19	50.14 ± 0.42
	Transported NCM	50.95 ± 0.25	86.94 ± 0.15	52.83 ± 0.18
	Radial only	50.93 ± 0.22	86.65 ± 0.10	52.88 ± 0.19
	ETF+NCM	51.47 ± 0.18	87.48 ± 0.12	53.26 ± 0.10
	ETF+raw radial	45.98 ± 1.03	87.09 ± 0.11	47.66 ± 0.96
	ETF+equal radius	51.47 ± 0.18	87.48 ± 0.12	53.26 ± 0.10
	ETF+no transport	48.65 ± 0.55	85.61 ± 0.18	50.14 ± 0.42
	Ours (RCR)	53.27 ± 0.34	87.74 ± 0.05	55.04 ± 0.21
Seq-TinyImageNet / 200	ETF only	25.77 ± 0.41	68.30 ± 0.62	28.58 ± 0.31
	Transported NCM	28.71 ± 0.33	68.69 ± 0.31	32.17 ± 0.37
	Radial only	28.21 ± 0.42	66.09 ± 1.00	32.14 ± 0.33
	ETF+NCM	29.29 ± 0.41	70.01 ± 0.35	32.63 ± 0.37
	ETF+raw radial	19.96 ± 2.18	54.13 ± 2.77	24.79 ± 1.02
	ETF+equal radius	29.29 ± 0.41	70.01 ± 0.35	32.63 ± 0.37
	ETF+no transport	25.77 ± 0.41	68.31 ± 0.68	28.58 ± 0.31
	Ours (RCR)	31.42 ± 0.65	69.59 ± 0.81	34.76 ± 0.56
Seq-TinyImageNet / 500	ETF only	26.61 ± 0.32	67.71 ± 0.39	29.37 ± 0.42
	Transported NCM	29.59 ± 0.37	69.37 ± 0.35	32.92 ± 0.24
	Radial only	30.00 ± 0.19	67.33 ± 0.13	33.72 ± 0.24
	ETF+NCM	29.95 ± 0.28	70.34 ± 0.30	33.27 ± 0.17
	ETF+raw radial	24.14 ± 0.90	58.26 ± 0.53	28.37 ± 0.66
	ETF+equal radius	29.95 ± 0.28	70.34 ± 0.30	33.27 ± 0.17
	ETF+no transport	26.62 ± 0.33	67.72 ± 0.41	29.38 ± 0.43
	Ours (RCR)	32.24 ± 0.04	70.16 ± 0.16	35.46 ± 0.07

Note: All heads use the same frozen features. No-transport uses the ETF vertex as the radial center, isolating the role of transported class measures.

Analysis. Table 5 uses the same frozen evaluation artifacts as the main results and separates ETF evidence, transported centroids, radial evidence, competitive radii, and transport. It reveals two failure modes. Transported NCM usually improves over ETF on hard routing regimes, so class representatives do benefit from being moved into the current feature frame. However, NCM alone does not supply a reliability model and can underperform on easier regimes. Conversely, ETF+no-transport collapses back to the ETF reference, showing that radial evidence only matters when it is attached to transported class measures. The full RCR row is strongest because it keeps the global ETF ordering while adding local class plausibility and competitive uncertainty.

G.2 Mixture Weight and Schedule

Table 6 shows that the fixed $q/(q+1)$ mixture is in a broad useful region rather than a hidden test-selected constant. Table 7 tests whether the deterministic stage schedule is necessary for the Seq-CIFAR-100 artifacts. The fixed-cap variant removes the stage ramp and is used by the main paper.

Table 6: **Sensitivity to the ETF/NC-shrunk radial mixture weight.**

Set	ω	Class	Task	Route
C10/200	0.00	66.14 ± 1.38	96.74 ± 0.10	66.58 ± 1.34
	0.25	67.49 ± 1.15	96.82 ± 0.09	67.93 ± 1.11
	0.50	69.09 ± 0.70	96.78 ± 0.11	69.56 ± 0.62
	0.75	70.72 ± 0.56	96.62 ± 0.08	71.19 ± 0.53
	Ours	70.36 ± 0.50	96.67 ± 0.08	70.82 ± 0.46
	1.00	67.09 ± 0.34	96.43 ± 0.14	67.62 ± 0.24
C10/500	0.00	72.16 ± 1.30	96.82 ± 0.12	72.71 ± 1.35
	0.25	73.53 ± 1.45	96.88 ± 0.10	74.09 ± 1.49
	0.50	73.81 ± 1.02	96.88 ± 0.04	74.42 ± 1.08
	0.75	73.48 ± 0.51	96.75 ± 0.02	74.15 ± 0.63
	Ours	73.68 ± 0.72	96.81 ± 0.03	74.33 ± 0.83
	1.00	70.82 ± 0.39	96.41 ± 0.01	71.52 ± 0.25
C100/200	0.00	41.71 ± 0.61	85.53 ± 0.16	42.81 ± 0.57
	0.25	43.48 ± 0.42	86.34 ± 0.20	44.63 ± 0.39
	0.50	45.76 ± 0.28	87.10 ± 0.15	47.01 ± 0.14
	0.75	48.33 ± 0.52	87.73 ± 0.22	49.74 ± 0.40
	Ours	49.74 ± 0.58	87.28 ± 0.20	51.43 ± 0.47
	1.00	48.08 ± 0.48	86.34 ± 0.20	49.98 ± 0.44
C100/500	0.00	48.65 ± 0.55	85.54 ± 0.19	50.14 ± 0.42
	0.25	49.90 ± 0.51	86.48 ± 0.15	51.46 ± 0.38
	0.50	51.19 ± 0.25	87.35 ± 0.10	52.76 ± 0.11
	0.75	52.86 ± 0.35	87.90 ± 0.07	54.53 ± 0.19
	Ours	53.27 ± 0.34	87.74 ± 0.05	55.04 ± 0.21
	1.00	51.72 ± 0.38	87.10 ± 0.08	53.58 ± 0.28
Tiny/200	0.00	25.77 ± 0.41	68.30 ± 0.62	28.58 ± 0.31
	0.25	26.82 ± 0.50	69.70 ± 0.63	29.65 ± 0.48
	0.50	28.16 ± 0.41	70.95 ± 0.41	31.15 ± 0.33
	0.75	30.28 ± 0.59	71.38 ± 0.50	33.29 ± 0.48
	Ours	31.42 ± 0.65	69.59 ± 0.81	34.76 ± 0.56
	1.00	30.38 ± 0.47	68.12 ± 0.72	34.03 ± 0.51
Tiny/500	0.00	26.61 ± 0.32	67.71 ± 0.39	29.37 ± 0.42
	0.25	27.80 ± 0.29	69.21 ± 0.51	30.62 ± 0.40
	0.50	29.23 ± 0.30	70.64 ± 0.50	32.12 ± 0.46
	0.75	31.31 ± 0.25	71.50 ± 0.32	34.30 ± 0.38
	Ours	32.24 ± 0.04	70.16 ± 0.16	35.46 ± 0.07
	1.00	31.44 ± 0.25	68.92 ± 0.12	34.90 ± 0.17

Note: Class, Task, and Route report final accuracies. The public method uses the deterministic fixed value $\omega = q/(q+1)$, not a value selected on test data.

Table 7: **Ablation of the RCR mixture schedule on Seq-CIFAR-100.** The deployed fixed-cap rule uses $\omega = q/(q+1)$ at every task, while the stage schedule uses $\omega_t = \min(q/(q+1), (t-1)/t)$. Both variants use the NC-shrunk radial evidence and are evaluated on the same frozen artifacts; lower FF is better.

Buffer	Mixture rule	n	Class FAA	Class FF	Task FAA	Route
200	Stage schedule	2	49.55	19.31	87.49	51.21
200	**Ours: fixed cap**	2	49.49	16.43	87.39	51.18
500	Stage schedule	2	53.12	11.44	87.72	54.91
500	**Ours: fixed cap**	2	53.13	10.29	87.71	54.94

Note: The fixed-cap rule removes the deterministic stage ramp and is used by the reported Ours head. It reaches nearly the same final routing as the staged variant on these runs, while applying maximum radial evidence from the first task.

Analysis. The mixture ablations rule out a narrow schedule artifact. A broad interval of ω remains competitive, and removing the deterministic stage schedule does not remove the gain. This is why the main method uses the simpler fixed cap $\omega = q/(q+1)$: it is tied to the number of classes per block rather than to a tuned task index, while preserving the Class-IL and route improvements. Even the global value $\omega = 0.5$ improves over ETF in every dataset/buffer cell in Table 6, so the gain is not created by a single test-selected mixture constant.

G.3 Radius and NC Retraction

Table 8 shows that the competitive radius exponent is not a narrow tuned constant; the default $\alpha_\rho = 0.25$ lies in a stable range. Table 9 isolates the NC-retraction clipping: the lower endpoint is the natural value 0, while the upper safeguard 0.95 avoids fully trusting transported centers after drift compensation.

Table 8: **Sensitivity to the competitive-radius exponent.** Seq-CIFAR-100 buffer 200, final task. The default $\alpha_\rho = 0.25$ is stable: larger changes in the exponent preserve the routing gain, while the exact value is not selected on the test set.

α_ρ	Class-IL ↑	Task-IL ↑	Route ↑	Wrong-task ↓
0.00	46.44 ± 0.46	87.06 ± 0.22	48.02 ± 0.45	51.98 ± 0.45
0.10	48.56 ± 0.43	87.15 ± 0.17	50.18 ± 0.35	49.82 ± 0.35
Ours: 0.25	49.74 ± 0.58	87.28 ± 0.20	51.43 ± 0.47	48.57 ± 0.47
0.50	48.54 ± 0.78	87.35 ± 0.21	50.27 ± 0.70	49.73 ± 0.70
1.00	43.61 ± 1.05	86.76 ± 0.36	45.34 ± 1.01	54.66 ± 1.01

Table 9: **Upper clipping for NC retraction.** We set $\alpha_{\text{lo}} = 0$ and compare the deployed upper safeguard $\alpha_{\text{hi}} = 0.95$ to the unconstrained endpoint $\alpha_{\text{hi}} = 1$. The lower clip is inactive on these runs; the useful safeguard is preventing full trust in transported centers.

Dataset / Buffer	Setting	α_{hi}	Class FAA ↑	Class FF ↓
Seq-CIFAR-10 / 200	No upper safeguard	1.00	70.44 ± 0.59	22.63 ± 1.14
Seq-CIFAR-10 / 200	**Ours**	0.95	70.36 ± 0.50	22.85 ± 1.19
Seq-CIFAR-10 / 500	No upper safeguard	1.00	73.56 ± 0.60	17.60 ± 0.72
Seq-CIFAR-10 / 500	**Ours**	0.95	73.68 ± 0.72	17.73 ± 0.72
Seq-CIFAR-100 / 200	No upper safeguard	1.00	49.41 ± 0.58	17.46 ± 0.51
Seq-CIFAR-100 / 200	**Ours**	0.95	49.74 ± 0.58	16.27 ± 0.37
Seq-CIFAR-100 / 500	No upper safeguard	1.00	53.06 ± 0.40	10.96 ± 0.28
Seq-CIFAR-100 / 500	**Ours**	0.95	53.27 ± 0.34	10.18 ± 0.31
Seq-TinyImageNet / 200	No upper safeguard	1.00	30.57 ± 0.58	26.46 ± 0.82
Seq-TinyImageNet / 200	**Ours**	0.95	31.42 ± 0.65	23.84 ± 0.94
Seq-TinyImageNet / 500	No upper safeguard	1.00	31.45 ± 0.12	25.12 ± 0.20
Seq-TinyImageNet / 500	**Ours**	0.95	32.24 ± 0.04	22.47 ± 0.10

Analysis. The radius and shrinkage tables show that RCR is not driven by a fragile radius constant. Moderate changes in α_ρ keep the same qualitative behavior: larger class uncertainty reduces radial confidence, and the ETF component preserves global comparability. The NC retraction mainly stabilizes forgetting and late-stage routing by preventing unreliable transported centers from moving too far away from their neural-collapse vertices.

H Replay Calibration Baselines

Replay-fitted calibrators are useful stress tests because they fit task or class biases from replay labels. They are not the same claim as RCR: RCR does not fit a task router from replay and does not add a task oracle. The comparison in Table 10 asks whether replay-side score calibration alone explains the gains. It does not: replay calibration can help in some

Table 10: **Replay-fitted calibration does not explain RCR.**

Dataset/Buffer	Head	Class-IL ↑	Task-IL ↑	Route ↑	Wrong-task ↓
Seq-CIFAR-10 / 200	ProNC ETF	66.14 ± 1.38	96.74 ± 0.10	66.58 ± 1.34	33.42 ± 1.34
	Replay task-bias	59.21 ± 0.65	96.74 ± 0.10	59.60 ± 0.64	40.40 ± 0.64
	Replay class-bias	59.36 ± 0.62	96.72 ± 0.12	59.75 ± 0.60	40.25 ± 0.60
	Replay task affine	58.19 ± 1.35	96.74 ± 0.10	58.61 ± 1.32	41.39 ± 1.32
	Ours (RCR)	70.36 ± 0.50	96.67 ± 0.08	70.82 ± 0.46	29.18 ± 0.46
Seq-CIFAR-10 / 500	ProNC ETF	72.16 ± 1.30	96.82 ± 0.12	72.71 ± 1.35	27.29 ± 1.35
	Replay task-bias	67.89 ± 0.91	96.82 ± 0.12	68.35 ± 0.97	31.65 ± 0.97
	Replay class-bias	68.03 ± 0.88	96.87 ± 0.07	68.49 ± 0.93	31.51 ± 0.93
	Replay task affine	67.84 ± 0.49	96.82 ± 0.12	68.33 ± 0.47	31.67 ± 0.47
	Ours (RCR)	73.68 ± 0.72	96.81 ± 0.03	74.33 ± 0.83	25.67 ± 0.83
Seq-CIFAR-100 / 200	ProNC ETF	41.71 ± 0.61	85.53 ± 0.16	42.81 ± 0.57	57.19 ± 0.57
	Replay task-bias	29.91 ± 0.63	85.53 ± 0.16	30.67 ± 0.59	69.33 ± 0.59
	Replay class-bias	29.36 ± 0.65	84.86 ± 0.06	30.13 ± 0.70	69.87 ± 0.70
	Replay task affine	27.54 ± 2.57	85.53 ± 0.16	28.35 ± 2.56	71.65 ± 2.56
	Ours (RCR)	49.74 ± 0.58	87.28 ± 0.20	51.43 ± 0.47	48.57 ± 0.47
Seq-CIFAR-100 / 500	ProNC ETF	48.65 ± 0.55	85.54 ± 0.19	50.14 ± 0.42	49.86 ± 0.42
	Replay task-bias	36.14 ± 0.85	85.54 ± 0.19	37.06 ± 0.91	62.94 ± 0.91
	Replay class-bias	36.22 ± 0.35	85.51 ± 0.36	37.12 ± 0.29	62.88 ± 0.29
	Replay task affine	36.03 ± 0.89	85.54 ± 0.19	37.05 ± 0.82	62.95 ± 0.82
	Ours (RCR)	53.27 ± 0.34	87.74 ± 0.05	55.04 ± 0.21	44.96 ± 0.21
Seq-TinyImageNet / 200	ProNC ETF	25.77 ± 0.41	68.30 ± 0.62	28.58 ± 0.31	71.42 ± 0.31
	Replay task-bias	28.48 ± 0.11	68.30 ± 0.62	31.43 ± 0.11	68.57 ± 0.11
	Replay class-bias	29.49 ± 0.15	72.05 ± 0.21	32.50 ± 0.14	67.50 ± 0.14
	Replay task affine	31.42 ± 0.45	68.30 ± 0.62	34.48 ± 0.40	65.52 ± 0.40
	Ours (RCR)	31.42 ± 0.65	69.59 ± 0.81	34.76 ± 0.56	65.24 ± 0.56
Seq-TinyImageNet / 500	ProNC ETF	26.61 ± 0.32	67.71 ± 0.39	29.37 ± 0.42	70.63 ± 0.42
	Replay task-bias	29.80 ± 0.16	67.71 ± 0.39	32.59 ± 0.17	67.41 ± 0.17
	Replay class-bias	30.37 ± 0.32	71.54 ± 0.25	33.28 ± 0.27	66.72 ± 0.27
	Replay task affine	32.02 ± 0.28	67.71 ± 0.39	35.14 ± 0.20	64.86 ± 0.20
	Ours (RCR)	32.24 ± 0.04	70.16 ± 0.16	35.46 ± 0.07	64.54 ± 0.07

Note: All calibration heads are fit only on replay-buffer scores and labels, then evaluated on held-out test features.

TinyImageNet cells, but it is not consistently competitive with RCR and can damage CIFAR routing.

Analysis. Replay calibration is a useful negative control. If routing drift were only a global task-bias problem, replay-fitted offsets would match RCR. Instead, their behavior is inconsistent across datasets and memory budgets. This supports the main interpretation: the useful signal is not a learned task router, but the combination of transported class evidence and class-specific reliability under the same ETF decision space.

I Mechanistic Diagnostics

The main paper argues that RCR acts when learned class measures deviate from the symmetric ETF geometry. The diagnostics below make this claim explicit through task-by-task curves, score-profile examples, stage-level NC proxies, route oracle bounds, and class-wise correlations.

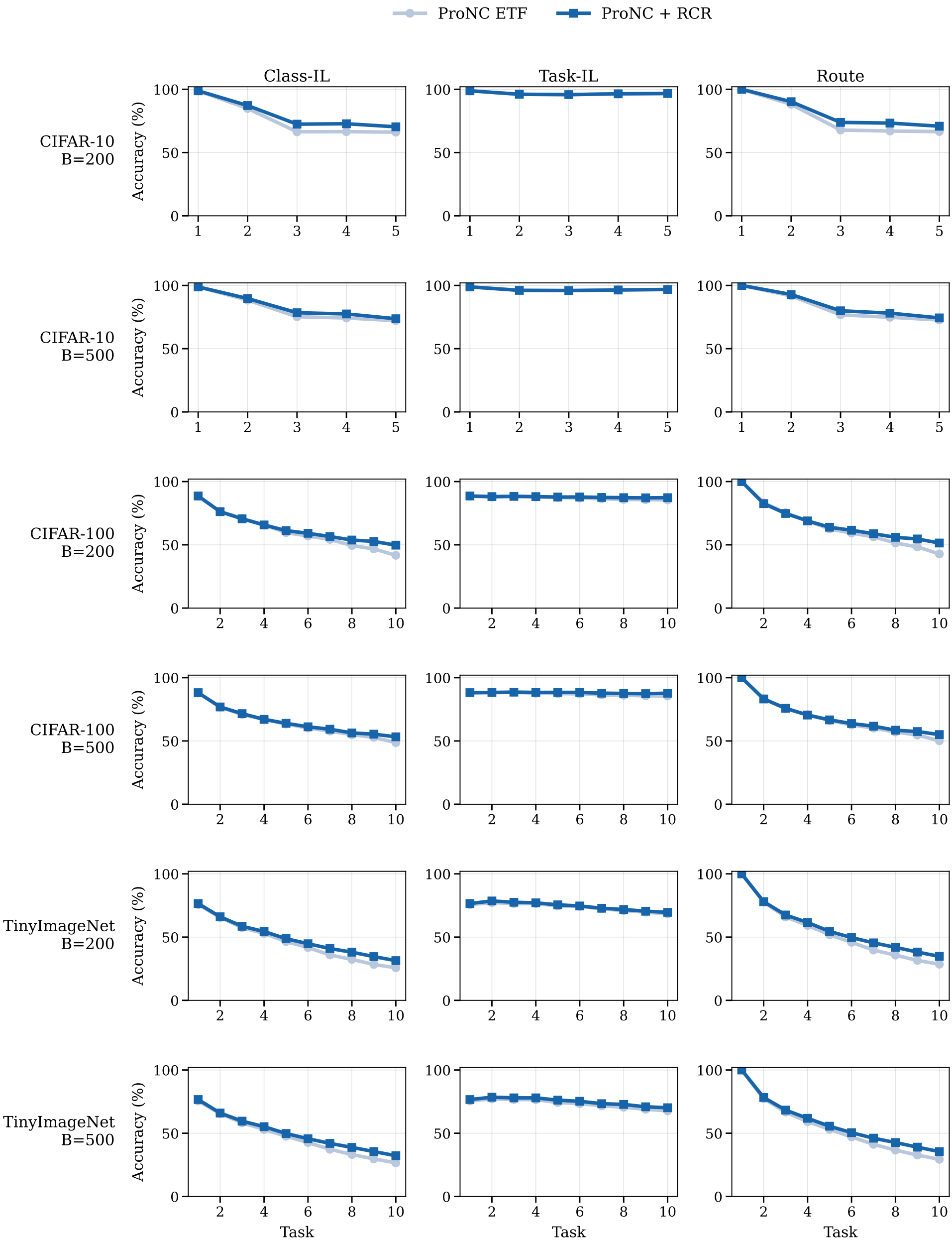

Figure 1: **Task-by-task diagnostic curves across all validated datasets.** Each row shows one dataset and memory budget; columns report Class-IL accuracy, Task-IL accuracy, and route accuracy over the continual sequence. The same frozen checkpoints are evaluated with the ProNC ETF head and ProNC + RCR. Across datasets, RCR primarily changes the Class-IL and Route curves while leaving Task-IL close to the reference head, supporting the routing-drift interpretation.

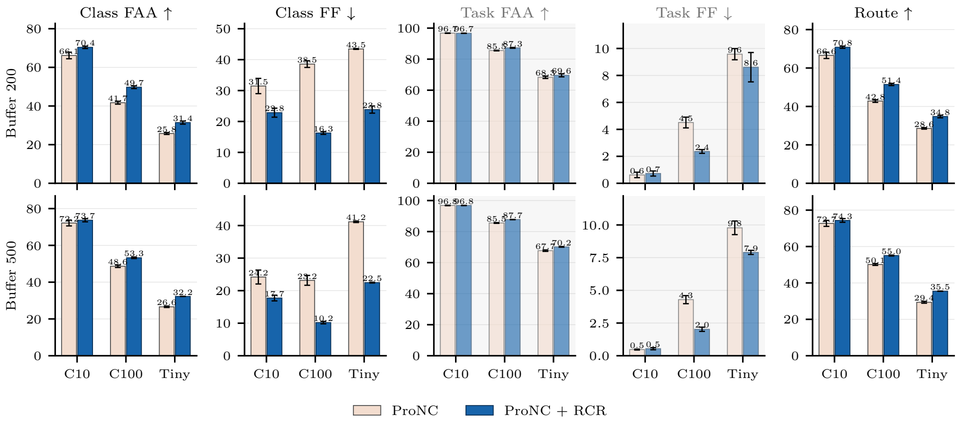

Figure 2: **Final head comparison across datasets and memory budgets.** Each panel compares the ProNC ETF head with ProNC +RCR on the same saved evaluation artifacts. Rows correspond to buffer sizes 200 and 500; columns report Class-IL FAA, Class-IL FF, Task-IL FAA, Task-IL FF, and final route accuracy.

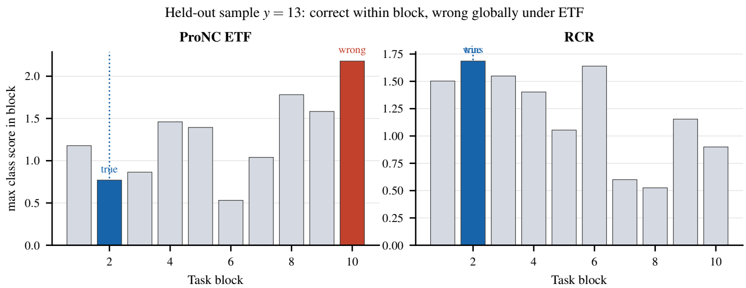

Figure 3: **A routing error at the block-score level.** For one held-out Seq-CIFAR-100 sample, the true class wins within its own task block, but the ProNC ETF maximum is larger in a wrong block. RCR changes the cross-block score profile so the correct block wins, without using task identity at test time.

Analysis. The diagnostic curves and block-score example visualize the same mechanism as the tables. RCR changes Class-IL and route accuracy together, while Task-IL is largely preserved. The block-score example makes this concrete: the sample is not a within-task class failure, because the correct class wins inside its own block; the error is caused by cross-block score comparability, which the reliability-calibrated score changes.

Table 11: **Neural-collapse deviations and routing drift on Seq-CIFAR-100, buffer 200.** Panel (a) tracks stage-level NC diagnostics and held-out performance. Panel (b) reports final-stage class-wise Pearson correlations: prototype–ETF deviation and competitive radius identify older classes with larger route error and larger RCR gains, while NC1 dispersion mostly tracks age. Values are mean ± std over saved runs.

(a) Stage-level diagnostics

Stage	$\langle p,u\rangle\uparrow$	NC1 disp. ↓	CV(ρ)	Route ↑	Class-IL ↑
T3	0.39 ± 0.00	0.27 ± 0.00	0.08 ± 0.01	$75.34\pm1.01\to74.79\pm0.83$	$70.98\pm0.89\to70.58\pm0.63$
T10	0.19 ± 0.00	0.41 ± 0.00	0.11 ± 0.00	$42.81\pm0.57\to51.43\pm0.47$	$41.71\pm0.61\to49.74\pm0.58$

(b) Final-stage class-wise correlations

Class signal	Route error	ΔClass	Age
$\delta_c=d_{\cos}(p_c,u_c)$	0.68 ± 0.02	0.59 ± 0.03	0.90 ± 0.02
ρ_c	0.61 ± 0.00	0.55 ± 0.05	0.94 ± 0.01
NC1 disp.	0.24 ± 0.03	0.13 ± 0.04	0.81 ± 0.01

Note: NC1 dispersion is the held-out mean $d_{\cos}(z,p_c)$ within each class. Route error and ΔClass are computed per class before correlation.

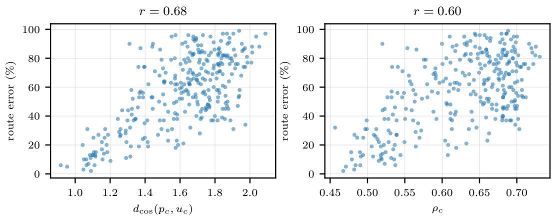

Figure 4: **Class-wise NC-deviation signals predict routing error.** On Seq-CIFAR-100 with buffer 200, classes whose transported prototypes deviate further from their ETF vertices, or whose competitive radii are larger, tend to have larger held-out route error under the ProNC ETF head.

Table 12: **Route-oracle bounds.**

Dataset/Buffer	Head	Class-IL ↑	Task-IL ↑	Route ↑
Seq-CIFAR-10 / 200	ProNC ETF	66.14 ± 1.38	96.74 ± 0.10	66.58 ± 1.34
	RCR	70.36 ± 0.50	96.67 ± 0.08	70.82 ± 0.46
	ProNC + true route	96.74 ± 0.10	96.74 ± 0.10	100.00 ± 0.00
	RCR + true route	96.67 ± 0.08	96.67 ± 0.08	100.00 ± 0.00
Seq-CIFAR-10 / 500	ProNC ETF	72.16 ± 1.30	96.82 ± 0.12	72.71 ± 1.35
	RCR	73.68 ± 0.72	96.81 ± 0.03	74.33 ± 0.83
	ProNC + true route	96.82 ± 0.12	96.82 ± 0.12	100.00 ± 0.00
	RCR + true route	96.81 ± 0.03	96.81 ± 0.03	100.00 ± 0.00
Seq-CIFAR-100 / 200	ProNC ETF	41.71 ± 0.61	85.53 ± 0.16	42.81 ± 0.57
	RCR	49.74 ± 0.58	87.28 ± 0.20	51.43 ± 0.47
	ProNC + true route	85.53 ± 0.16	85.53 ± 0.16	100.00 ± 0.00
	RCR + true route	87.28 ± 0.20	87.28 ± 0.20	100.00 ± 0.00
Seq-CIFAR-100 / 500	ProNC ETF	48.65 ± 0.55	85.54 ± 0.19	50.14 ± 0.42
	RCR	53.27 ± 0.34	87.74 ± 0.05	55.04 ± 0.21
	ProNC + true route	85.54 ± 0.19	85.54 ± 0.19	100.00 ± 0.00
	RCR + true route	87.74 ± 0.05	87.74 ± 0.05	100.00 ± 0.00
Seq-TinyImageNet / 200	ProNC ETF	25.77 ± 0.41	68.30 ± 0.62	28.58 ± 0.31
	RCR	31.42 ± 0.65	69.59 ± 0.81	34.76 ± 0.56
	ProNC + true route	68.30 ± 0.62	68.30 ± 0.62	100.00 ± 0.00
	RCR + true route	69.59 ± 0.81	69.59 ± 0.81	100.00 ± 0.00
Seq-TinyImageNet / 500	ProNC ETF	26.61 ± 0.32	67.71 ± 0.39	29.37 ± 0.42
	RCR	32.24 ± 0.04	70.16 ± 0.16	35.46 ± 0.07
	ProNC + true route	67.71 ± 0.39	67.71 ± 0.39	100.00 ± 0.00
	RCR + true route	70.16 ± 0.16	70.16 ± 0.16	100.00 ± 0.00

Note: The true-route rows restrict the argmax to the ground-truth task block and are upper bounds, not deployable methods.

Table 13: **Class-wise NC-deviation correlations.**

Dataset/Buffer	Signal	Route error r	ΔClass r
Seq-CIFAR-10 / 200	$d_{\cos}(p_c, u_c)$	0.76 ± 0.11	0.39 ± 0.16
	ρ_c	0.76 ± 0.16	0.36 ± 0.17
	NC1 disp.	0.70 ± 0.11	0.33 ± 0.11
	Age	0.45 ± 0.26	0.31 ± 0.32
Seq-CIFAR-10 / 500	$d_{\cos}(p_c, u_c)$	0.72 ± 0.13	0.49 ± 0.17
	ρ_c	0.73 ± 0.16	0.49 ± 0.17
	NC1 disp.	0.67 ± 0.15	0.39 ± 0.19
	Age	0.42 ± 0.29	0.54 ± 0.20
Seq-CIFAR-100 / 200	$d_{\cos}(p_c, u_c)$	0.68 ± 0.02	0.59 ± 0.03
	ρ_c	0.61 ± 0.00	0.55 ± 0.05
	NC1 disp.	0.24 ± 0.03	0.13 ± 0.04
	Age	0.46 ± 0.01	0.38 ± 0.05
Seq-CIFAR-100 / 500	$d_{\cos}(p_c, u_c)$	0.61 ± 0.01	0.62 ± 0.02
	ρ_c	0.46 ± 0.02	0.55 ± 0.02
	NC1 disp.	0.21 ± 0.00	0.14 ± 0.02
	Age	0.27 ± 0.02	0.36 ± 0.02
Seq-TinyImageNet / 200	$d_{\cos}(p_c, u_c)$	0.76 ± 0.01	0.41 ± 0.05
	ρ_c	0.50 ± 0.03	0.43 ± 0.04
	NC1 disp.	0.44 ± 0.02	0.08 ± 0.02
	Age	0.57 ± 0.04	0.16 ± 0.07
Seq-TinyImageNet / 500	$d_{\cos}(p_c, u_c)$	0.71 ± 0.00	0.44 ± 0.01
	ρ_c	0.54 ± 0.03	0.42 ± 0.06
	NC1 disp.	0.47 ± 0.01	0.11 ± 0.03
	Age	0.54 ± 0.02	0.21 ± 0.02

Note: Pearson correlations are computed per seed across classes and then averaged.

Analysis. Table 11 reports stage-level NC diagnostics and class-wise correlations, while Table 12 reports the true-route upper bound. Together they show that routing, not task-local discrimination, is the dominant remaining headroom. Finally, Table 13 and Figure 4 connect the effect to neural-collapse deviations at class level: classes with larger prototype-to-ETF deviation or larger competitive radius tend to have larger route error, which is exactly where radial reliability should matter.

References

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